Gradient Descent (m & b)

step-1 – Initialize random values of m & b

$m = 1$ and $b = 0$

Step-2 – Decide epochs & learning rate

epochs = 100 , $lr = 0.01$

For i in epochs:

$b = b_{old} - \eta \, slope$

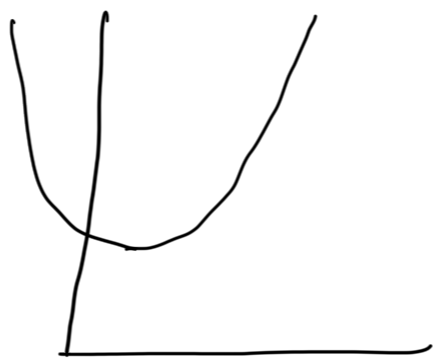
$m = m_{old} - \eta \, slope$

Slope calculation

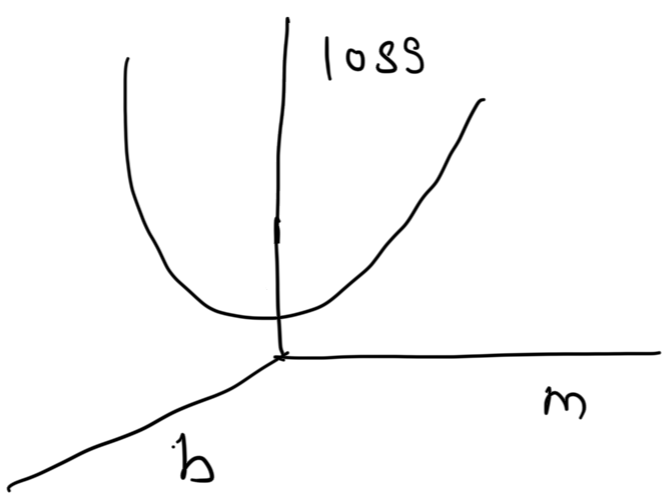
We know, $L = \sum (y - \hat{y})^2$

$$= \sum (y - mx - b)^2$$

In case of only b loss function was parabolic (calculated last time by considering only value of b)



This time we are considering both m & b so loss Function becomes



$$L(m,b) = \sum (y_i - mx_i - b)^2$$

Partial derivative need to apply

$$b_slope = \frac{\partial L}{\partial b}_{\text{at } b=0} = -2\sum(y_i - mx_i - b)$$

$$m_slope = \frac{\partial L}{\partial m}_{\text{at } m=1}$$

$$= -2\sum(y_i - mx - b)x_i$$